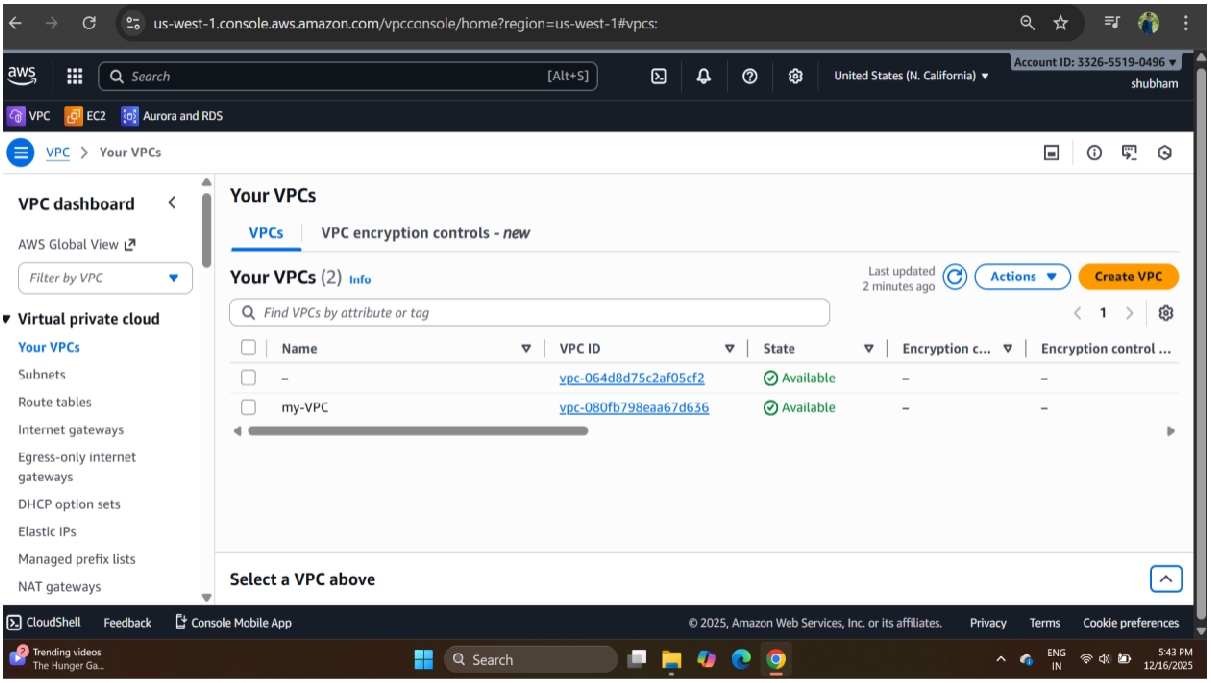
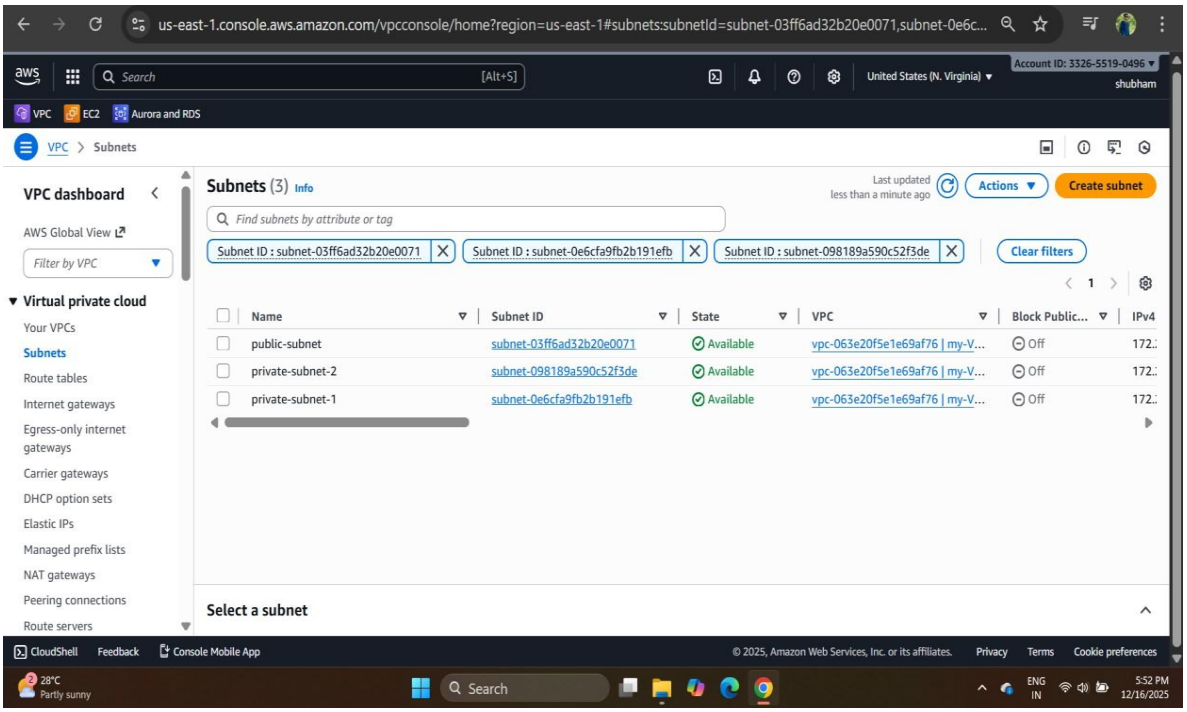
TASK : 3- Tier Project (Tomcat & Mysql)

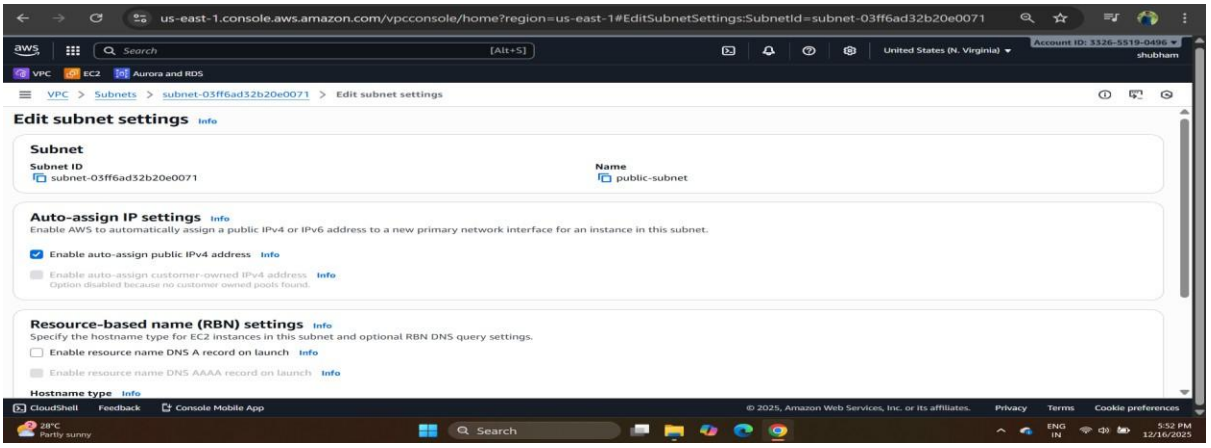
1. Create VPC.



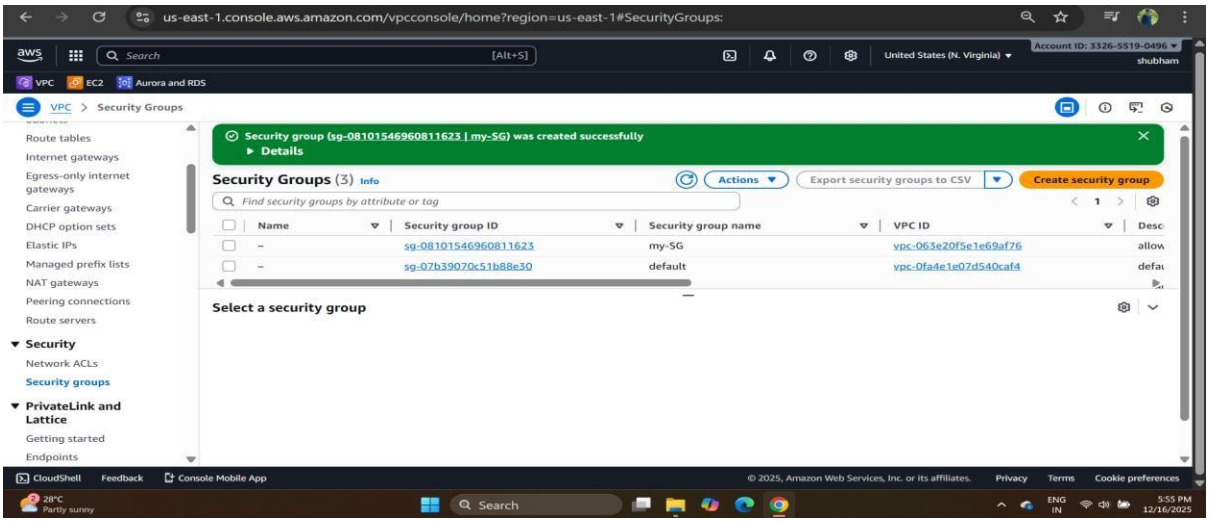
2. Create 3 subnets of this VPC (Public-subnet,private-subnet-1,private-subnet-2)



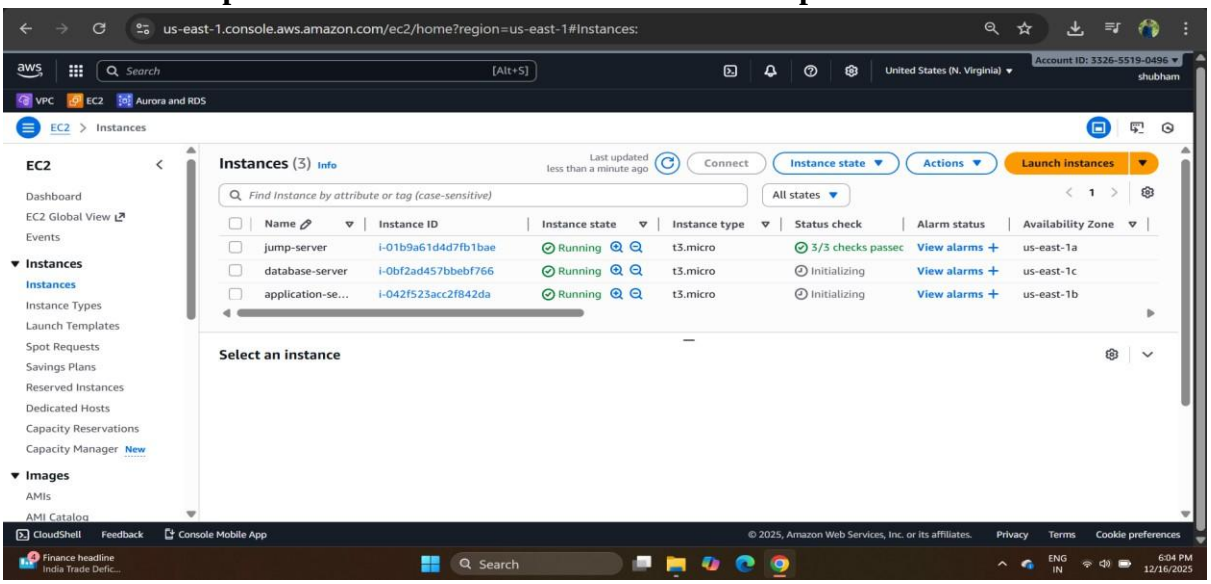
3. **Make public subnet by using auto-assign IP**



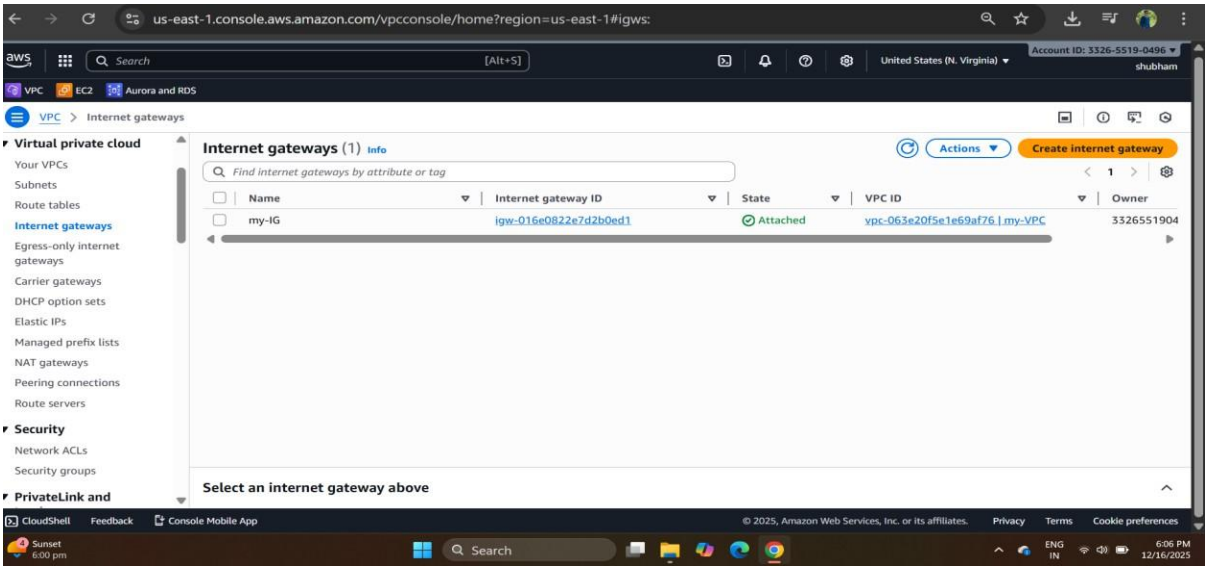
4. **Now create security group allow SSH,Tomcat,Mysql,HTTP.**



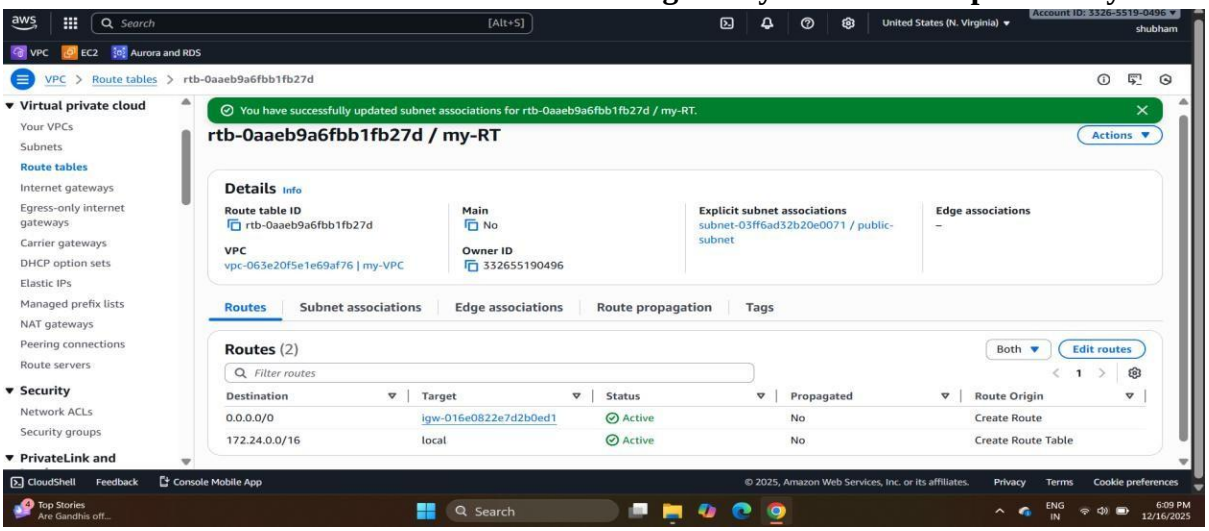
5. create instances in subnets **1.Jump server create in the public subnet 2.Application server create in private subnet 3.database server create in private subnet.**



6. Now create **Internet gateway and attach to VPC.**



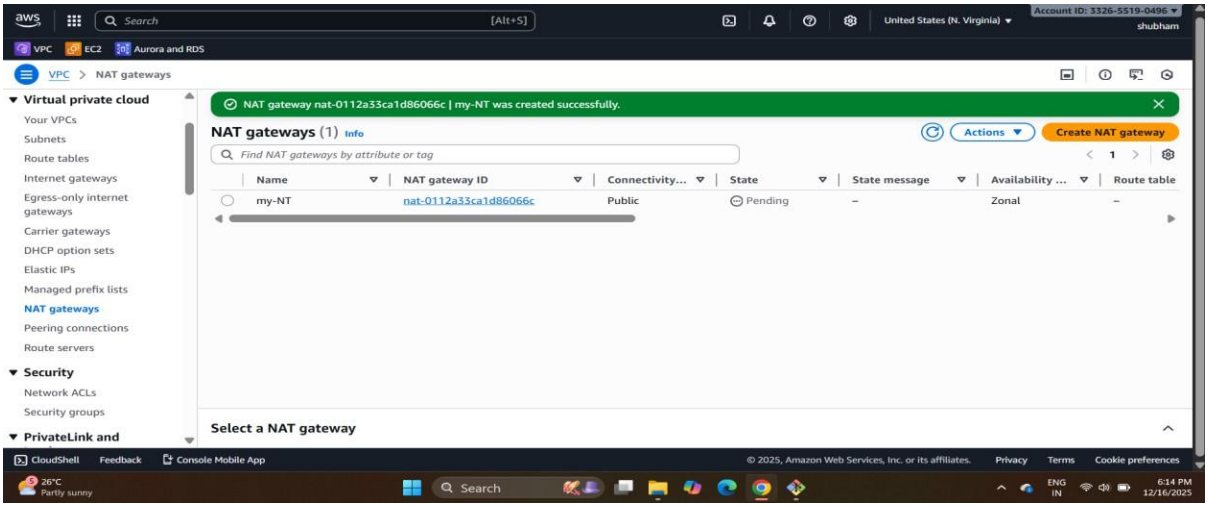
7. Now create **Route table and edit route internet gateway and associate public only.**



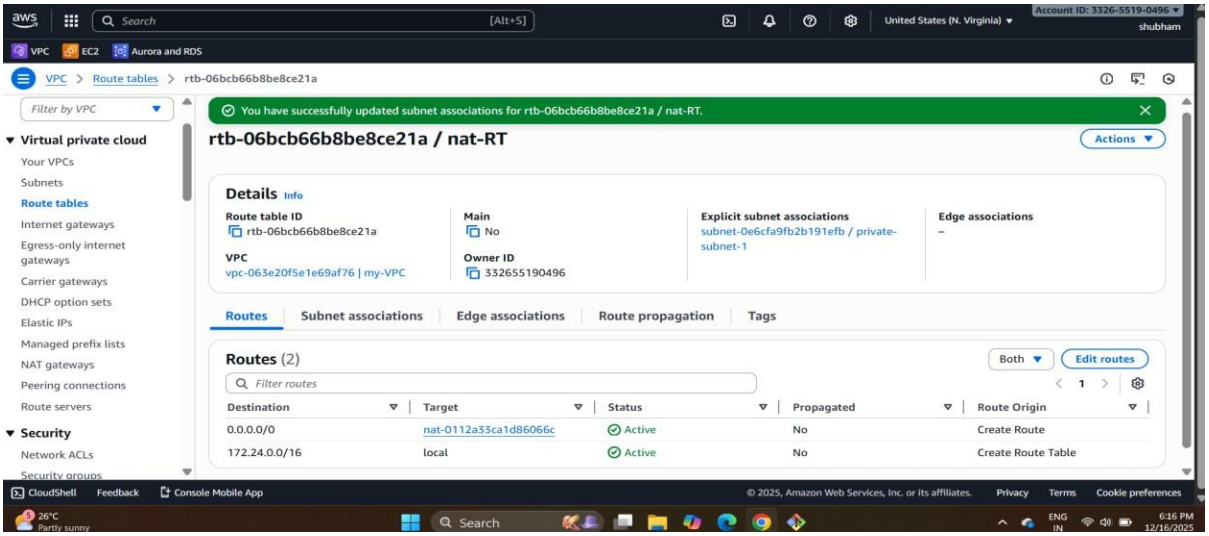
8. Tansfer key pair to jump server **get access of application server.**

```
hp@shubham MINGW64 ~
$ cd Downloads/

hp@shubham MINGW64 ~/Downloads
$ scp -i new.key.pem new.key.pem ec2-user@3.94.144.44:/home/ec2-user
The authenticity of host '3.94.144.44 (3.94.144.44)' can't be established.
ED25519 key fingerprint is: SHA256:eXZ1U/mv1OTKxOa/Z5R8lNlxcZ/3o6N4jhR5OX3r9co
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '3.94.144.44' (ED25519) to the list of known hosts.
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
new.key.pem                                   100% 1678     5.5KB/s   00:00

hp@shubham MINGW64 ~/Downloads
$ |
```

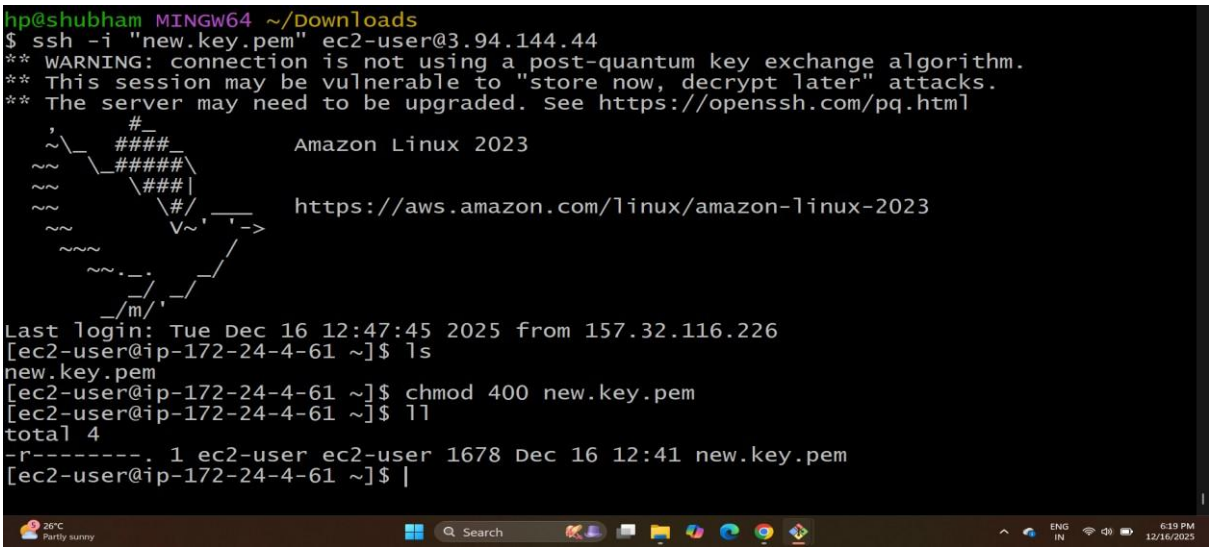
9. After access application server to need internet so create **NAT gateway in public subnet.**



10. After this create route table for **NAT gateway associate to private subnet edit route NAT gateway and associate private subnet-1.**



11. Access the jump serevr and **change the key permission and get access of application serevr.**



12. Download **apache Tomcat** in application serevr.

```
[ec2-user@ip-172-24-19-9 ~]$ curl -O https://dlcdn.apache.org/tomcat/tomcat-9/v9.0.113
/bin/apache-tomcat-9.0.113.tar.gz
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 12.4M  100 12.4M    0     0  27.7M      0 --:--:-- --:--:-- --:--:-- 27.7M
[ec2-user@ip-172-24-19-9 ~]$ ls
apache-tomcat-9.0.113.tar.gz
[ec2-user@ip-172-24-19-9 ~]$
```

13. Extract this file in **/opt folder.**

```
[ec2-user@ip-172-24-19-9 ~]$ sudo tar -xzvf apache-tomcat-9.0.113.tar.gz -C /opt
apache-tomcat-9.0.113/conf/
apache-tomcat-9.0.113/conf/catalina.policy
apache-tomcat-9.0.113/conf/catalina.properties
apache-tomcat-9.0.113/conf/context.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xml
apache-tomcat-9.0.113/conf/jaspic-providers.xsd
apache-tomcat-9.0.113/conf/logging.properties
apache-tomcat-9.0.113/conf/server.xml
apache-tomcat-9.0.113/conf/tomcat-users.xml
apache-tomcat-9.0.113/conf/tomcat-users.xsd
apache-tomcat-9.0.113/conf/web.xml
apache-tomcat-9.0.113/bin/
apache-tomcat-9.0.113/lib/
apache-tomcat-9.0.113/logs/
apache-tomcat-9.0.113/temp/
apache-tomcat-9.0.113/webapps/
apache-tomcat-9.0.113/webapps/ROOT/
apache-tomcat-9.0.113/webapps/ROOT/WEB-INF/
apache-tomcat-9.0.113/webapps/docs/
apache-tomcat-9.0.113/webapps/docs/META-INF/
apache-tomcat-9.0.113/webapps/docs/WEB-INF/
apache-tomcat-9.0.113/webapps/docs/WEB-INF/jsp/
apache-tomcat-9.0.113/webapps/docs/annotationapi/
apache-tomcat-9.0.113/webapps/docs/api/
```

14. Go to **/opt/Apache tomcat-9.0.113/** and in that bin/ here you see catellina.sh so we have to start tomcat but tomcat need java so we **first install java.**

```
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# cd bin/
[root@ip-172-24-19-9 bin]# ls
bootstrap.jar                configtest.sh     shutdown.sh
catalina-tasks.xml           daemon.sh         startup.bat
catalina.bat                 digest.bat        startup.sh
catalina.sh                  digest.sh         tomcat-juli.jar
ciphers.bat                  makebase.bat      tomcat-native.tar.gz
ciphers.sh                   makebase.sh       tool-wrapper.bat
commons-daemon-native.tar.gz setclasspath.bat  tool-wrapper.sh
commons-daemon.jar           setclasspath.sh   version.bat
configtest.bat               shutdown.bat      version.sh
[root@ip-172-24-19-9 bin]# yum install java -y
Amazon Linux 2023 repository                        69 MB/s |  51 MB     00:00
Amazon Linux 2023 Kernel Livepatch repository      247 kB/s |  29 kB     00:00
Dependencies resolved.
=================================================================================
 Package                        Arch   Version                    Repository   Size
=================================================================================
Installing:
 java-25-amazon-corretto        x86_64 1:25.0.1+9-1.amzn2023.1    amazonlinux 206 k
Installing dependencies:
 alsa-lib                       x86_64 1.2.7.2-1.amzn2023.0.2     amazonlinux 504 k
 cairo                          x86_64 1.18.0-4.amzn2023.0.3      amazonlinux 717 k
 dejavu-sans-fonts              noarch 2.37-16.amzn2023.0.2       amazonlinux 1.3 M
```

15. Now start tomcat **./catellina.sh**

```
[root@ip-172-24-19-9 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat
-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-24-19-9 bin]#
```

16. Now we have to dump website in application server so go to **/opt/apache tomcat-9.0.113/webapps/.**

```
[root@ip-172-24-19-9 ~]# cd /opt/apache-tomcat-9.0.113/webapps/
[root@ip-172-24-19-9 webapps]# ls
ROOT  docs  examples  host-manager  manager
[root@ip-172-24-19-9 webapps]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-c
it/student.war
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100 89423  100 89423    0     0   214k      0 --:--:-- --:--:-- --:--:--  214k
[root@ip-172-24-19-9 webapps]#
```

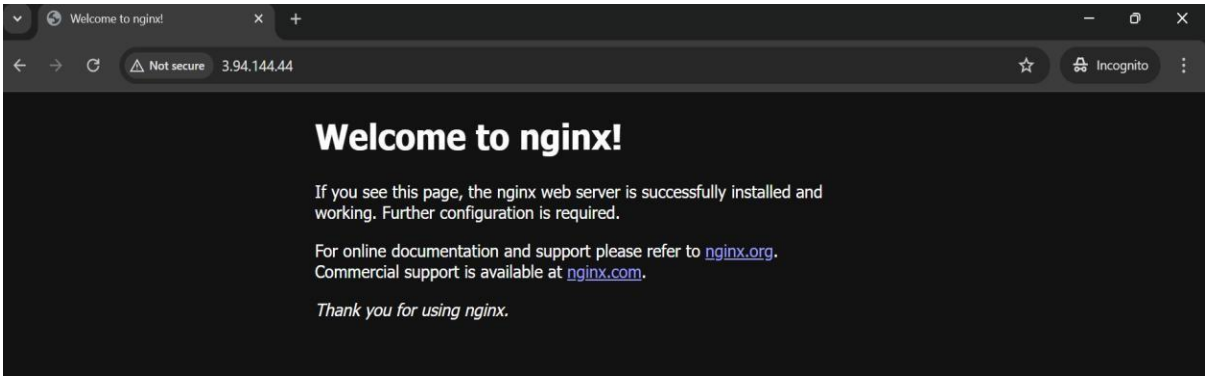
17. Now everything set but we have to give location to jump server but we need nginx so **install nginx in jump server.**

```
[root@ip-172-24-4-61 ~]# yum install nginx -y
Amazon Linux 2023 repository                          70 MB/s |  51 MB     00:00
Amazon Linux 2023 Kernel Livepatch repository        260 kB/s |  29 kB     00:00
Dependencies resolved.
=================================================================================
 Package                 Arch       Version                     Repository      Size
=================================================================================
Installing:
 nginx                   x86_64     1:1.28.0-1.amzn2023.0.2     amazonlinux     33 k
Installing dependencies:
 generic-logos-httpd     noarch     18.0.0-12.amzn2023.0.3      amazonlinux     19 k
 gperftools-libs         x86_64     2.9.1-1.amzn2023.0.3        amazonlinux    308 k
 libunwind               x86_64     1.4.0-5.amzn2023.0.3        amazonlinux     66 k
 nginx-core              x86_64     1:1.28.0-1.amzn2023.0.2     amazonlinux    686 k
 nginx-filesystem        noarch     1:1.28.0-1.amzn2023.0.2     amazonlinux    9.6 k
 nginx-mimetypes         noarch     2.1.49-3.amzn2023.0.3       amazonlinux     21 k

Transaction Summary
=================================================================================
Install  7 Packages

Total download size: 1.1 M
Installed size: 3.7 M
Downloading Packages:
(1/7): generic-logos-httpd-18.0.0-12.amzn2023.0.3.noa 513 kB/s |  19 kB     00:00
```

18. Start & enable nginx.service.

```
[root@ip-172-24-4-61 ~]# systemctl start nginx.service
[root@ip-172-24-4-61 ~]# systemctl enable nginx.service
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/s
ystemd/system/nginx.service.
[root@ip-172-24-4-61 ~]#
```

19. Then hit the public IP address of jump server you see nginx page.



After this we have to add location of application server to see website.

19. First login root of application server & go to **/etc/nginx/ here you see nginx.conf so in this file we have add location.**

```
[root@ip-172-24-4-61 ~]# cd /etc/nginx/
[root@ip-172-24-4-61 nginx]# ls
conf.d                  koi-utf             scgi_params
default.d               koi-win             scgi_params.default
fastcgi.conf            mime.types          uwsgi_params
fastcgi.conf.default    mime.types.default  uwsgi_params.default
fastcgi_params          nginx.conf          win-utf
fastcgi_params.default  nginx.conf.default
[root@ip-172-24-4-61 nginx]# vim nginx.conf
```

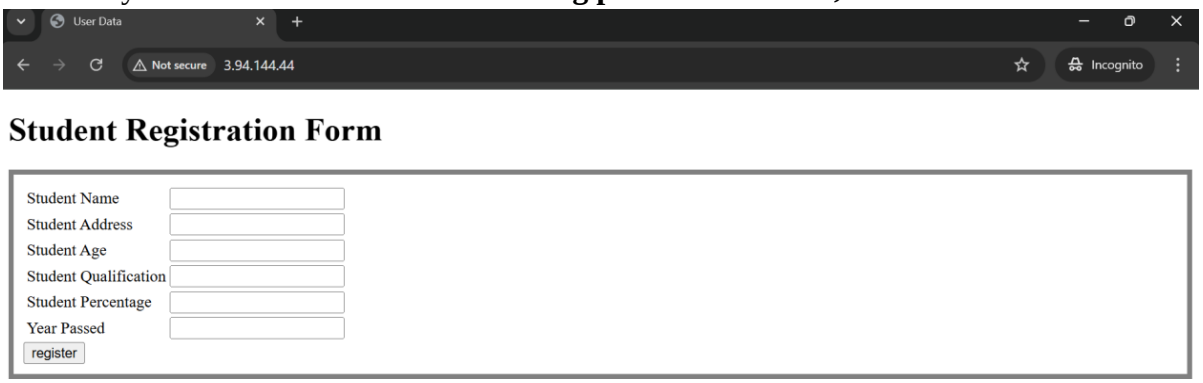
In this location we need to give application server of private IP address with folder name to identify the folder & website.

```
        # Load configuration files for the default server block.
        include /etc/nginx/default.d/*.conf;

        error_page 404 /404.html;
        location = /404.html {
        }
        location / {
        proxy_pass http://172.24.19.9:8080/student/;
        }
        error_page 500 502 503 504 /50x.html;
        location = /50x.html {
        }
    }

-- INSERT --                                                    50,38         53%
```

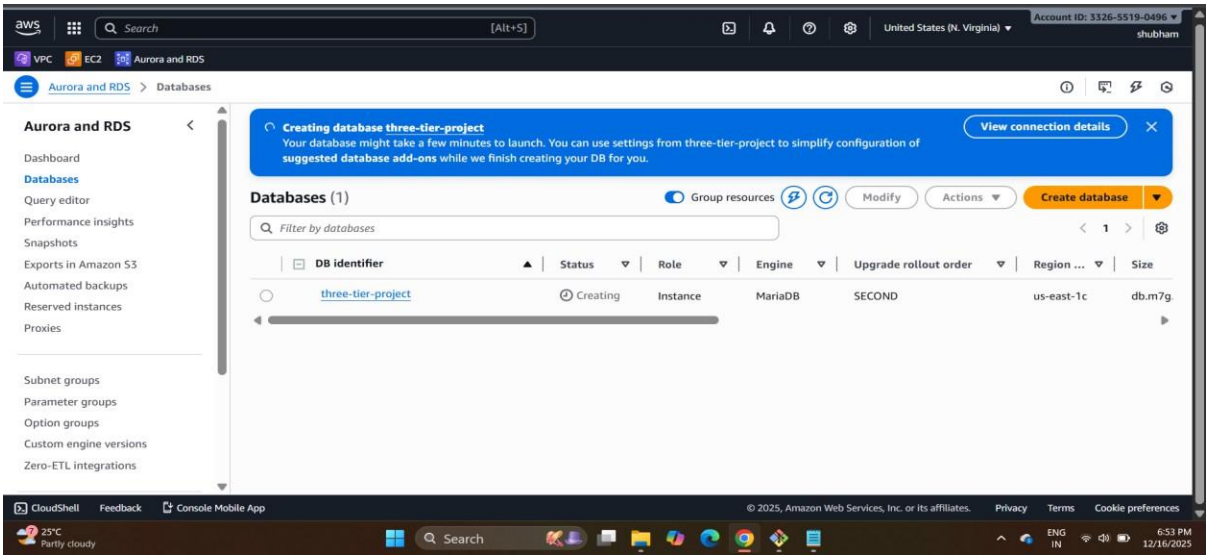
20. Then restart the nginx.service.

```
[root@ip-172-24-4-61 nginx]# vim nginx.conf
[root@ip-172-24-4-61 nginx]# systemctl restart nginx.service
[root@ip-172-24-4-61 nginx]#
```

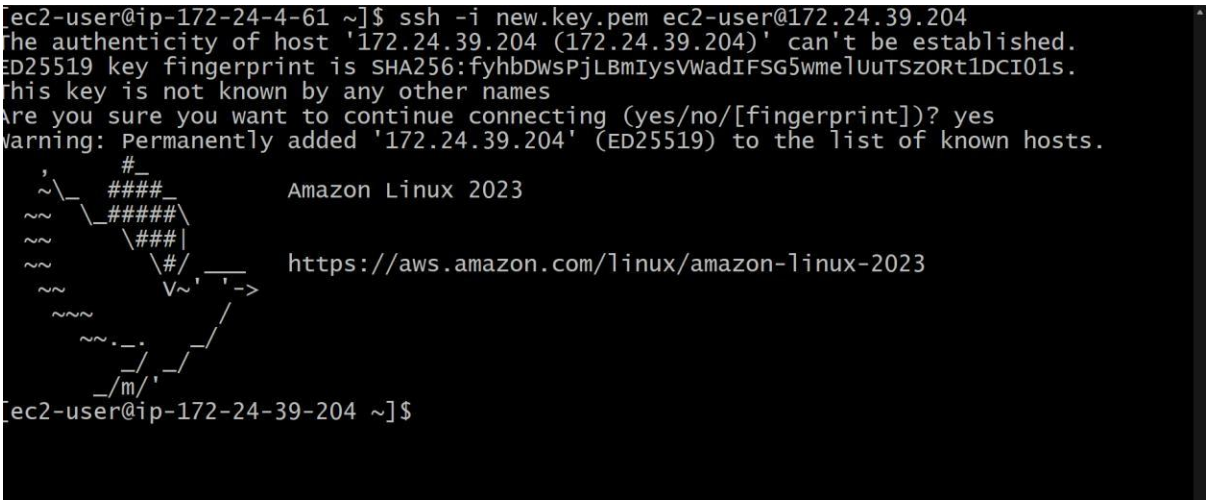
21. Now you can see the website after **hitting public IP address,but data not store.**



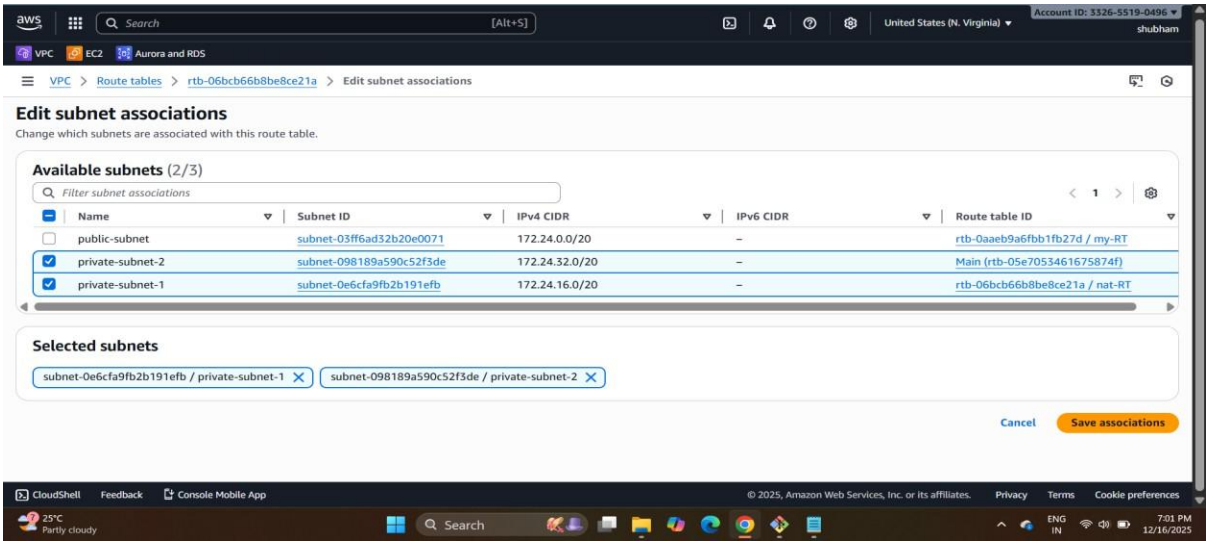
22. To store the data we create RDS database when create RDS select mariadb,username &manage password &also select owner VPC &security group but select avalibilty zone select zone when you create database server & create RDS database instance there.



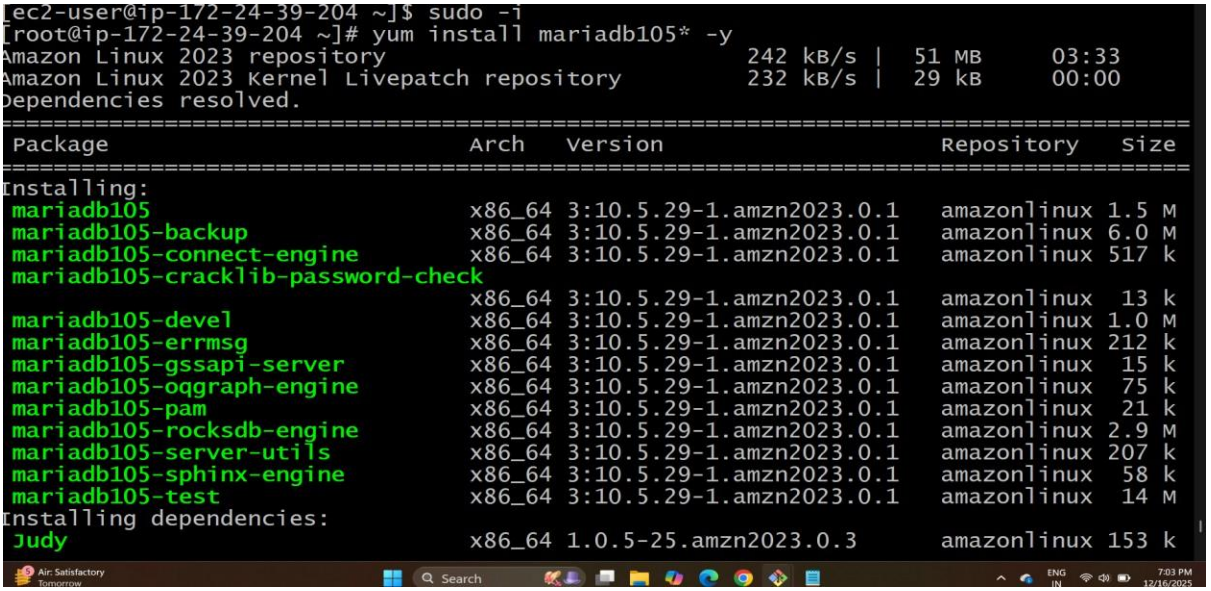
23. Get the **access of database instance.**



24. Install mariadb105 to provide internet to database instance go to **nat-rt & associate private-subnet-2.**



25. Now install **mariadb 105***



26. Start & enable **mariadb.service**

27. After create RDS database instance you get access RDS instance in database server – **mysql –h <endpoint> -u <username> -p<password>. & also configure the database & create table for website command.**

```
onaws.com
[root@ip-172-24-39-204 ~]# mysql -h three-tier-project.c0h22oq66o6a.us-east-1.rds.amaz
onaws.com -u shubham -pshubham1234
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MariaDB connection id is 34
Server version: 11.4.8-MariaDB-log managed by https://aws.amazon.com/rds/

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MariaDB [(none)]> create database studentapp;
Query OK, 1 row affected (0.003 sec)

MariaDB [(none)]> use studentapp;
Database changed
MariaDB [studentapp]> CREATE TABLE if not exists students(student_id INT NOT NULL AUTO_INCREMENT,
    -> student_name VARCHAR(100) NOT NULL,
    ->     student_addr VARCHAR(100) NOT NULL,
    -> student_age VARCHAR(3) NOT NULL,
    -> student_qual VARCHAR(20) NOT NULL,
    -> student_percent VARCHAR(10) NOT NULL,
    -> student_year_passed VARCHAR(10) NOT NULL,
    -> PRIMARY KEY (student_id)
    -> );
Query OK, 0 rows affected (0.009 sec)

MariaDB [studentapp]> show tables;
+----------------------+
| Tables_in_studentapp |
+----------------------+
| students             |
+----------------------+
1 row in set (0.001 sec)

MariaDB [studentapp]>
```

28. Now log out database server & log in application server to connect application database .After login root of app go to - /opt/apache tomcat 9.0.113/lib/

Here we have download mysql connector to connect the database.

```
root@ip-172-24-19-9:/opt/apache-tomcat-9.0.113/lib
[root@ip-172-24-19-9 ~]# cd /opt/apache-tomcat-9.0.113/
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# ls
BUILDING.txt     LICENSE  README.md      RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE   RELEASE-NOTES  bin          lib   temp  work
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# cd lib/
[root@ip-172-24-19-9 lib]# ls
annotations-api.jar       ecj-4.20.jar     tomcat-api.jar         tomcat-i18n-fr.jar     tomcat-jni.jar
catalina-ant.jar          el-api.jar       tomcat-coyote-ffm.jar  tomcat-i18n-ja.jar     tomcat-util-scan.jar
catalina-ha.jar           jasper-el.jar    tomcat-coyote.jar      tomcat-i18n-ko.jar     tomcat-util.jar
catalina-ssi.jar          jasper.jar       tomcat-dbcp.jar        tomcat-i18n-pt-BR.jar  tomcat-websocket.jar
catalina-storeconfig.jar  jaspic-api.jar   tomcat-i18n-cs.jar     tomcat-i18n-ru.jar     websocket-api.jar
catalina-tribes.jar       jsp-api.jar      tomcat-i18n-de.jar     tomcat-i18n-zh-CN.jar
catalina.jar              servlet-api.jar  tomcat-i18n-es.jar     tomcat-jdbc.jar
[root@ip-172-24-19-9 lib]# curl -O https://s3-us-west-2.amazonaws.com/studentapi-cit/mysql-connector.jar
  % Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
                                 Dload  Upload   Total   Spent    Left  Speed
100  983k  100  983k    0     0   873k      0  0:00:01  0:00:01 --:--:--  873k
[root@ip-172-24-19-9 lib]#
```

29. Now we have add location of database in application server. Go to- /opt/apache tomcat 9.0.113/conf/

Here you see context.xml so in this we add location & save it.

```
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# ls
BUILDING.txt     LICENSE  README.md      RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE   RELEASE-NOTES  bin          lib   temp  work
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# cd conf/
[root@ip-172-24-19-9 conf]# ls
Catalina        catalina.properties  jaspic-providers.xml  logging.properties  tomcat-users.xml  web.xml
catalina.policy  context.xml         jaspic-providers.xsd  server.xml          tomcat-users.xsd
[root@ip-172-24-19-9 conf]# vim context.xml
```

In this file we **edit username & password of database & also the endpoint of RDS instance.**

```
     http://www.apache.org/licenses/LICENSE-2.0

  Unless required by applicable law or agreed to in writing, software
  distributed under the License is distributed on an "AS IS" BASIS,
  WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
  See the License for the specific language governing permissions and
  limitations under the License.
-->
<!-- The contents of this file will be loaded for each web application -->
<Context>

    <!-- Default set of monitored resources. If one of these changes, the    -->
    <!-- web application will be reloaded.                                   -->
    <WatchedResource>WEB-INF/web.xml</WatchedResource>
    <WatchedResource>WEB-INF/tomcat-web.xml</WatchedResource>
    <WatchedResource>${catalina.base}/conf/web.xml</WatchedResource>

    <!-- Uncomment this to disable session persistence across Tomcat restarts -->
    <!--
    <Manager pathname="" />
    -->
    <Resource name="jdbc/TestDB" auth="Container" type="javax.sql.DataSource"
    maxTotal="500" maxIdle="30" maxWaitMillis="1000"
    username="shubham" password="shubham1234" driverClassName="com.mysql.jdbc.Driver"
    url="jdbc:mysql://three-tier-project.c0h22oq66o6a.us-east-1.rds.amazonaws.co:3306/studentapp?useUnicode=yes&amp;characterE
ncoding=utf8"/>


</Context>
                                                                    34,80          Bot
```

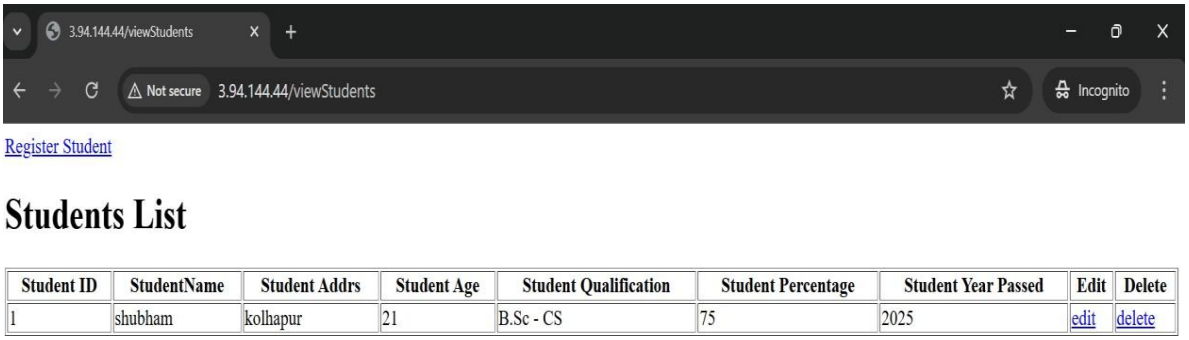
30. Then stop the **catellina.sh & again start it.**

```
root@ip-172-24-19-9:/opt/apache-tomcat-9.0.113/bin
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# ls
BUILDING.txt     LICENSE  README.md      RUNNING.txt  conf  logs  webapps
CONTRIBUTING.md  NOTICE   RELEASE-NOTES  bin          lib   temp  work
[root@ip-172-24-19-9 apache-tomcat-9.0.113]# cd bin/
[root@ip-172-24-19-9 bin]# ./catalina.sh stop
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
NOTE: Picked up JDK_JAVA_OPTIONS:  --add-opens=java.base/java.lang=ALL-UNNAMED --add-opens=java.base/java.lang.invoke=ALL-UNNA
MED --add-opens=java.base/java.lang.reflect=ALL-UNNAMED --add-opens=java.base/java.io=ALL-UNNAMED --add-opens=java.base/java.u
til=ALL-UNNAMED --add-opens=java.base/java.util.concurrent=ALL-UNNAMED --add-opens=java.rmi/sun.rmi.transport=ALL-UNNAMED
[root@ip-172-24-19-9 bin]# ./catalina.sh start
Using CATALINA_BASE:   /opt/apache-tomcat-9.0.113
Using CATALINA_HOME:   /opt/apache-tomcat-9.0.113
Using CATALINA_TMPDIR: /opt/apache-tomcat-9.0.113/temp
Using JRE_HOME:        /usr
Using CLASSPATH:       /opt/apache-tomcat-9.0.113/bin/bootstrap.jar:/opt/apache-tomcat-9.0.113/bin/tomcat-juli.jar
Using CATALINA_OPTS:
Tomcat started.
[root@ip-172-24-19-9 bin]#
```

31. Then hitting the public IP address you can see the website.

After this all steps we create 3-Tier-project.

Here you can see the all data will be stored.



Register Student

Students List

Student ID	StudentName	Student Addrs	Student Age	Student Qualification	Student Percentage	Student Year Passed	Edit	Delete
1	shubham	kolhapur	21	B.Sc - CS	75	2025	edit	delete